BYUNGWOO JEON

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RESEARCH INTEREST

The north star of my research is to develop a unified and scalable vision framework aligned with multi-modalities and human preferences. To this end, I am delving into multi-modal learning (*e.g.*, VLM, UMM) and robot learning (*e.g.*, VLA, WAM), with a focus on *3D/4D world modeling* [P6] that bridges perception, cognition and action for real-world embodied intelligence. In parallel, I am exploring multi-modal systems that enable *recursive self-improvement*, allowing models or agents to iteratively refine their own capabilities through reasoning and interaction.

- **Multi-modal learning**
  - Recursive self-improvement [C5, C2]
  - World modeling [P6]
- **Robot learning**
  - Agentic robotics
  - Long-horizon planning [P3, W1]
- **High-dimensional vision**
  - Spatio-temporal understanding [C4, C3, C1]
  - Neural rendering

PUBLICATIONS

C: conference, J: journal, W: workshop, P: preprint / $*$ equal contribution, † equal advising

[C5] Latent Reasoning in Continuous Space for Unified Multimodal Models
Byungwoo Jeon, Yoonwoo Jeong, Hyunseok Lee, Minsu Cho†, Jinwoo Shin†
*Conference on Neural Information Processing Systems (**NeurIPS**), 2026*

[C4] Physically Plausible Multi-View Object Swapping for Robot Data Augmentation
Jonghoon Lee*, Seong Hyeon Park*, Byungwoo Jeon, Minha Lee, Jinwoo Shin
*Conference on Neural Information Processing Systems (**NeurIPS**), 2026*

[C3] SpatialBoost: Enhancing Visual Representation through Language-Guided Reasoning
Byungwoo Jeon*, Dongyoung Kim*, Huiwon Jang, Insoo Kim, Jinwoo Shin
*European Conference on Computer Vision (**ECCV**), 2026*

[C2] Vision-aligned Latent Reasoning for Multi-modal Large Language Model
Byungwoo Jeon, Yoonwoo Jeong, Hyunseok Lee, Minsu Cho†, Jinwoo Shin†
*International Conference on Machine Learning (**ICML**), 2026*

[C1] TrackIME: Enhanced Video Point Tracking via Instance Motion Estimation
Seong Hyeon Park, Huiwon Jang, Byungwoo Jeon, Sukmin Yun, Paul Hongsuck Seo, Jinwoo Shin
*Conference on Neural Information Processing Systems (**NeurIPS**), 2024, Spotlight Presentation (326/15671=2.08%)*

[P6] Learning the 4D World through Point Trajectory Forecasting
Byungwoo Jeon*, Sunghwan Hong*, Dohyeon Kim, Jonghoon Lee, Jungwoo Park, Inhee Lee, Doosan Baek, Jinil Kim, Taehyun Lim, Hanbyul Joo, Marc Pollefeys, Saining Xie†, Jinwoo Shin†
Under review

[P5] Let Vision-Language Models Generate Images through Their Representations
JB Kim, Byungwoo Jeon, Kyungmin Lee, Hyeonmo Kang, Cheonbok Park, Shinyoung Joo, Jinwoo Shin†, Jeonghoon Kim†
Under review

[P4] LayCraft: Crafting Layout Patterns from Language Instructions and Design Rules
Kiyoung Kwon, Byungwoo Jeon, Kunwoong Kim, Junhoo Lee, Minhyun Lee, Seong-Jin Park, Jinwoo Shin
Under review

[P3] When Is a Subtask Complete? Consistency-Aware Subtask Annotation for Hierarchical Robot Policies
Byungjun Yoon, Hyeonwoo Ma, Byungwoo Jeon, Jinwoo Shin
Under review

[P2] FORGE3D: Faithful Observation-Grounded 3D Mesh Generation via Latent Forcing
Jonghoon Lee*, Seong Hyeon Park*, Byungwoo Jeon, Doosan Baek, Jinil Kim, Jinwoo Shin
Under review

[P1] Cog3DMap: Multi-View Vision-Language Reasoning with 3D Cognitive Maps
Chanyoung Gwak*, Yoonwoo Jeong*, Byungwoo Jeon, Hyunseok Lee, Jinwoo Shin†, Minsu Cho†
Under review

[W1] Hierarchical Vision Language Action Model Using Success and Failure Demonstrations
Jeongeun Park, Jihwan Yoon, Byungwoo Jeon, Juhan Park, Jinwoo Shin, Kyungjae Lee, Namhoon Cho, Sangdoo Yun, Sungjoon Choi
CoRL Workshop on Safe and Robust Robot Learning for Operation in the Real World, 2025

EDUCATION

KAIST, Ph.D. student in Artificial Intelligence (advisor: Prof. Jinwoo Shin) — Sep 2024 - Current
Korea University, B.S. in Computer Science and Engineering, Statistics — Mar 2020 - Aug 2024

WORK EXPERIENCE

NYU Courant, Visiting Scholar (advisor: Prof. Saining Xie) — TBD
ALINLAB, Research Intern (advisor: Prof. Jinwoo Shin) — Mar 2023 - Aug 2024
Arcreal, ML Engineer (host: Prof. Jinwoo Shin) — Jan 2023 - Jan 2024
MLVLAB, Research Intern (advisor: Prof. Hyunwoo J. Kim) — Jul 2022 - Dec 2022
M-monstar, Fullstack Engineer — Jul 2021 - Aug 2021

EXTERNAL COLLABORATOR

NYU Courant, Working on 4D world modeling [P6] (advisor: Prof. Saining Xie) — Aug 2026 - Current
ETH-Zürich CVG, Working on 4D world modeling [P6] (advisor: Dr. Sunghwan Hong) — Jul 2026 - Current
POSTECH CVLAB, Working on VLM reasoning [C5, C2, P1] (advisor: Prof. Minsu Cho) — Apr 2025 - Sep 2026
Korea Univ. RILAB, Working on VLA reasoning [W1] (advisor: Prof. Sungjoon Choi) — Jul 2025 - Sep 2026

PROJECTS

NVIDIA, Latent World Models (2026)
RLWRLD, RLDX: Vision-Language-Action Model for Dexterous Manipulation (2026–)
SAMSUNG, Semiconductor Pattern Generation [P4] (2025–2027)

ACADEMIC SERVICES

Conference Reviewer: ICLR'27, ECCV'26
Workshop Reviewer: Space in Vision, Language, and Embodied AI (SpaVLE) @ NeurIPS'25

HONORS & AWARDS

Silver Prize, K-Data Science Hackathon, 2023 ($2,000)
Top 8, Artificial Intelligence Grand Challenge, 2022
Dean's List, Korea University, 2020–2023

EXTRACURRICULAR ACTIVITIES

Google Developer Groups on Campus (GDGoC) @ Korea University, Lead and Founder — Aug 2022 - Jul 2023

SELECTED TALKS

Korea Univ., GDGoC, "Meta Runner on AI," Mar 2026
Yonsei & Korea Univ., GDGoC, "Towards the New World of Computing: CV on Real-world," Dec 2024
Korea Univ., GDGoC, "Recent Advances in Machine Learning," Nov 2023

MENTORING

- **Byungjun Yoon**, Ph.D. student at KAIST AI — Mar 2025 - Current
  Topic: Hierarchical VLA [P3]
- **Jonghoon Lee**, M.S. student at KAIST AI — Apr 2025 - Current
  Topic: 3D/4D mesh generation [C4, P2]
- **Minha Lee**, M.S. student at KAIST AI — Mar 2026 - Current
  Topic: Agentic world modeling
- **Kiyoung Kwon**, now senior undergrad at SNU AI — Jan 2026 - Current
  Topic: Agentic AI4Science [P4]
- **Hyunseo Yang**, now senior undergrad at SNU CSE — Jul 2026 - Current
  Topic: GUI agent with world modeling
- **Seoyeon Byun**, M.S. student at KAIST AI — Aug 2022 - Aug 2024
  Topic: 3D/4D neural rendering, **Collaborate**: GDGoC 3rd Lead
- **Seonghu Jeon**, M.S. student at KAIST AI — Mar 2022 - Aug 2024
  Topic: 3D/4D neural rendering, **Collaborate**: GDGoC 3rd AI Core

REFERENCE

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Saining Xie, Assistant Professor at NYU Courant, Co-founder at AMI Labs
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Seong Hyeon Park, Assistant Professor at Ajou University
E-mail: seonghyp@ajou.ac.kr

Sunghwan Hong, Postdoc Fellow at ETH-Zürich
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